
Comparative Corpus Analysis of Scientific Abstracts in AI, Machine Learning and NLP

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Abstract

Scientific abstracts in AI, machine learning, and NLP provide a compact record of how research priorities shift over time. We analyse a corpus of arXiv abstracts spanning 1991–2021 across three subfields (`cs.AI`, `cs.LG`, `cs.CL`) to understand how research focus and framing have evolved. We investigate two analytic axes: text representation (TF-IDF unigrams, n-gram TF-IDF, and BERT embeddings) and topic discovery method (keyword analysis, LDA, and BERTopic). Our findings show that TF-IDF and n-gram keyword analysis most reliably capture paradigm shifts, tracing a clear transition from symbolic AI in the 1990s to transformer-dominated research by 2015–2021. BERTopic reveals that the modern era is defined by thematic diversification rather than concentration around a single topic. LDA surfaces coherent latent themes but is sensitive to vocabulary drift across eras. BERT embeddings organise papers by topic rather than era, making them less suited for temporal drift analysis. Truncated SVD trajectory analysis confirms gradual but accelerating vocabulary change, with the sharpest shift occurring between 2007–2014 and 2015–2021.

1 Introduction

The volume of scientific literature in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) has grown at an extraordinary rate over the past three decades. As the field has matured, the problems researchers pursue, the vocabulary they use, and the way they frame their contributions have changed fundamentally. Tracking these changes at scale is difficult to do manually, but text analytics provides a principled means of surfacing trends across thousands of papers [1].

This project uses the arXiv dataset available via HuggingFace, which contains scientific abstracts annotated with metadata including title, author, and category. We focus on three subject categories: `cs.AI` (Artificial Intelligence), `cs.LG` (Machine Learning), and `cs.CL` (Computation and Language / NLP). The dataset was filtered to these three categories to reduce noise from unrelated fields. Publication year was extracted from each paper’s arXiv identifier, and abstracts were grouped into four periods: 1991–1998, 1999–2006, 2007–2014, and 2015–2021. These boundaries broadly align with known transitions in the field: the symbolic AI era, the statistical learning era, the rise of deep learning, and the transformer era respectively.

To illustrate the vocabulary shifts present in the corpus, consider this abstract from the earliest period: “[PASTE ONE SENTENCE FROM A 1991–1998 ABSTRACT HERE – e.g. *something about expert systems, logic, or knowledge representation*].” Contrast this with a recent abstract from 2015–2021: “[PASTE ONE SENTENCE FROM A 2015–2021 ABSTRACT HERE – e.g. *something about pre-trained transformers, attention, or deep neural networks*].” Even at a glance, the differences in vocabulary and framing are substantial, motivating a systematic analysis.

A good solution to this task should satisfy three criteria. First, findings should be *consistent* — trends seen in one method should appear across others, not be an artefact of a single analytical choice. Second, outputs should be *interpretable* to a reader with domain knowledge, with keywords and topics corresponding to recognisable research concepts. Third, findings should *align with known patterns*, such as the emergence of deep learning after 2012 and transformer-based methods after 2017 [2].

To understand how different analytical choices affect these criteria, we explore two experimental axes. The first is **text representation**: we compare TF-IDF unigrams, n-gram TF-IDF (bigrams and trigrams), and contextual BERT embeddings. The second is **topic discovery method**: we compare TF-IDF keyword analysis, Latent Dirichlet Allocation (LDA), and BERTopic. We additionally apply Truncated SVD to trace the vocabulary trajectory of the field and t-SNE to visualise how BERT embeddings distribute across periods. Section 2 describes the pipeline. Section 3 presents results and evaluation. We conclude in Section 4.

2 Methods

2.1 Data Preparation

The corpus was loaded from the arXiv HuggingFace dataset [1] and filtered to `cs.AI`, `cs.LG`, and `cs.CL`. Unnecessary columns were removed, abstracts with missing text were dropped, and category labels were standardised. Papers were then assigned to one of four time periods: 1991–1998, 1999–2006, 2007–2014, and 2015–2021. A pronounced class imbalance exists across periods: Period 4 contains far more papers than earlier periods, reflecting the exponential growth of the field after 2012. To ensure fair comparisons, all keyword and topic scores were normalised rather than using raw counts.

2.2 Text Preprocessing

All abstracts went through the following pipeline: lowercasing, removal of special characters and digits, tokenisation, stopwords removal using the combined scikit-learn and NLTK English stopwords lists [3, 4], and lemmatisation to reduce tokens to their root forms. A custom domain stopwords list was additionally applied to suppress high-frequency but uninformative academic terms such as “paper”, “proposed”, and “results”, which would otherwise dominate TF-IDF scores without providing era-discriminative signal.

2.3 Analytic Axis 1: Text Representation

We compare three text representations, each offering a different trade-off between interpretability and semantic richness.

TF-IDF Unigrams (Baseline). TF-IDF weights each word by its frequency in a document relative to the corpus, producing a sparse vector where each dimension maps directly to a human-readable word. *Hypothesis: Unigram TF-IDF will identify distinctive keywords per period but will miss multi-word methodological phrases and semantically related terms.*

N-gram TF-IDF. We extend TF-IDF to bigrams and trigrams (range 2–3) using CountVectorizer [3], capturing compound phrases like “convolutional neural network” that unigram analysis splits. N-gram density is also tracked as a proxy for the technical specificity of writing over time. *Hypothesis: N-gram TF-IDF will reveal method-specific vocabulary shifts more clearly than unigrams, at the cost of a larger, sparser feature space.*

BERT Embeddings. We use `bert-base-uncased` [2] to produce dense 768-dimensional embeddings per abstract, computed as the mean-pooled final hidden layer over all tokens. *Hypothesis: BERT will capture semantic similarity effectively but may be less sensitive to the vocabulary-level paradigm shifts that characterise era transitions.*

2.4 Analytic Axis 2: Topic Discovery Method

TF-IDF Keyword Analysis (Baseline). For each period and category we extract the top- k terms by normalised TF-IDF score at both unigram and n-gram levels, to identify concepts gaining or losing prominence. *Hypothesis: Keyword analysis will be the most interpretable approach but may miss latent themes spanning multiple vocabulary items.*

Latent Dirichlet Allocation (LDA). LDA [5] models each document as a mixture of latent topics and each topic as a word distribution. We apply it to the preprocessed corpus to discover recurring thematic clusters without requiring labelled data, then track topic prevalence across periods. *Hypothesis: LDA will complement keyword analysis by grouping co-occurring terms into coherent themes, though its bag-of-words assumption may limit sensitivity to rapidly evolving technical vocabulary.*

BERTopic. BERTopic [6] clusters documents by their BERT embeddings and labels each cluster using class-based TF-IDF (c-TF-IDF). Topic intensity — the relative frequency of a topic within a period and category — is tracked over time. *Hypothesis: BERTopic will produce more semantically coherent topics than LDA, particularly for concepts that share meaning but vary in surface form across eras.*

2.5 Temporal Drift Visualisation

Truncated SVD [3] reduces the TF-IDF matrix to 2D; period centroids are connected to trace the field’s macroscopic vocabulary trajectory, with axis labels derived from the most extreme terms at each pole. **t-SNE** [7] is applied to BERT embeddings on a random sample of 5,000 abstracts (perplexity 45) to assess whether papers cluster by time period in semantic space.

3 Experimental Evaluation

3.1 Metrics and Setup

We evaluate each approach against three criteria: **consistency** (do findings replicate across methods?), **interpretability** (are outputs understandable to a domain reader?), and **alignment** (do findings match known milestones such as the rise of deep learning post-2012?). Interpretability and alignment are assessed qualitatively; we mitigate subjectivity by cross-referencing findings across all methods. Table 1 summarises the relative strengths of each approach.

All preprocessing artefacts were saved and reused across experiments to guarantee consistent inputs. TF-IDF and n-gram features used `scikit-learn` [3]. BERT embeddings used `bert-base-uncased` via `HuggingFace` [2]. BERTopic used the `bertopic` library [6].

Table 1: Qualitative comparison of all methods against evaluation criteria. ✓ = strong, ~ = moderate, × = weak.

Method	Consistency	Interpretability	Alignment
TF-IDF Unigrams	✓	✓	✓
N-gram TF-IDF	✓	✓	✓
BERT Embeddings	×	×	×
LDA	~	~	~
BERTopic	✓	~	✓
SVD Trajectory	✓	~	✓

3.2 Corpus Growth and Data Imbalance

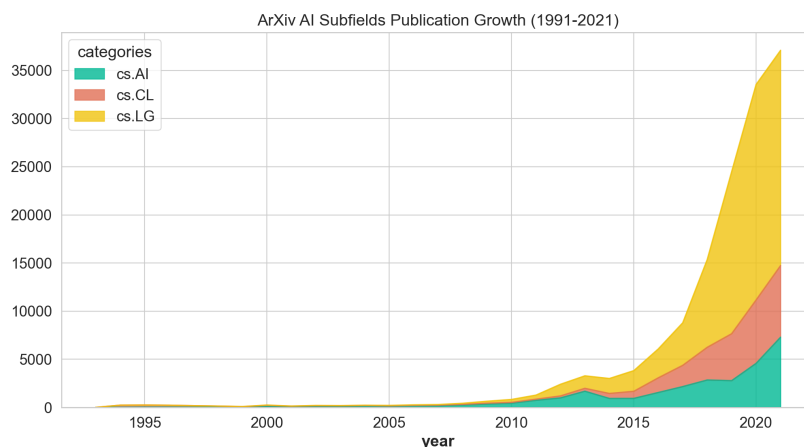


Figure 1: ArXiv AI subfield publication growth (1991–2021). cs.LG dominates from 2015 onward, reaching approximately 37,000 papers annually by 2021.

Figure 1 shows near-flat publication output until around 2010, followed by exponential growth driven primarily by cs.LG. This confirms the severe class imbalance across periods: Period 4 dwarfs earlier eras by orders of magnitude. Since raw counts would be dominated entirely by recent papers, all keyword and topic scores are normalised throughout the analysis.

3.3 Axis 1 Results: N-gram Key Terms per Period

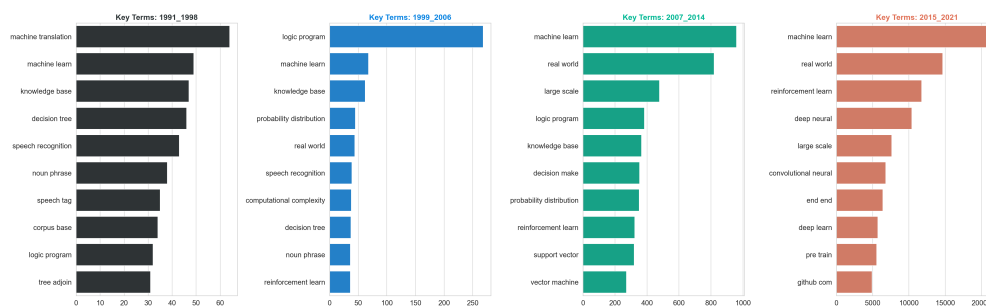


Figure 2: Top 10 bigram and trigram key terms per time period from n-gram TF-IDF analysis.

Figure 2 provides the clearest view of how research language evolved. In 1991–1998, dominant phrases include *machine translation*, *knowledge base*, *decision tree*, and *speech recognition* — the vocabulary of symbolic AI and early NLP. By 1999–2006, *logic program* becomes the single dominant phrase, suggesting a blend of symbolic and probabilistic approaches. In 2007–2014, *machine learn* takes over alongside *support vector* and *reinforcement learn*, marking the turn to statistical methods. By 2015–2021, *machine learn* is overwhelmingly frequent (count > 15,000), joined by *deep neural*, *convolutional neural*, *pre train*, and *reinforcement learn* — the language of the deep learning era. This confirms our hypothesis: n-grams surface compound methodological terms invisible to unigram analysis.

3.4 Axis 1 Results: Keyword Evolution Heatmap

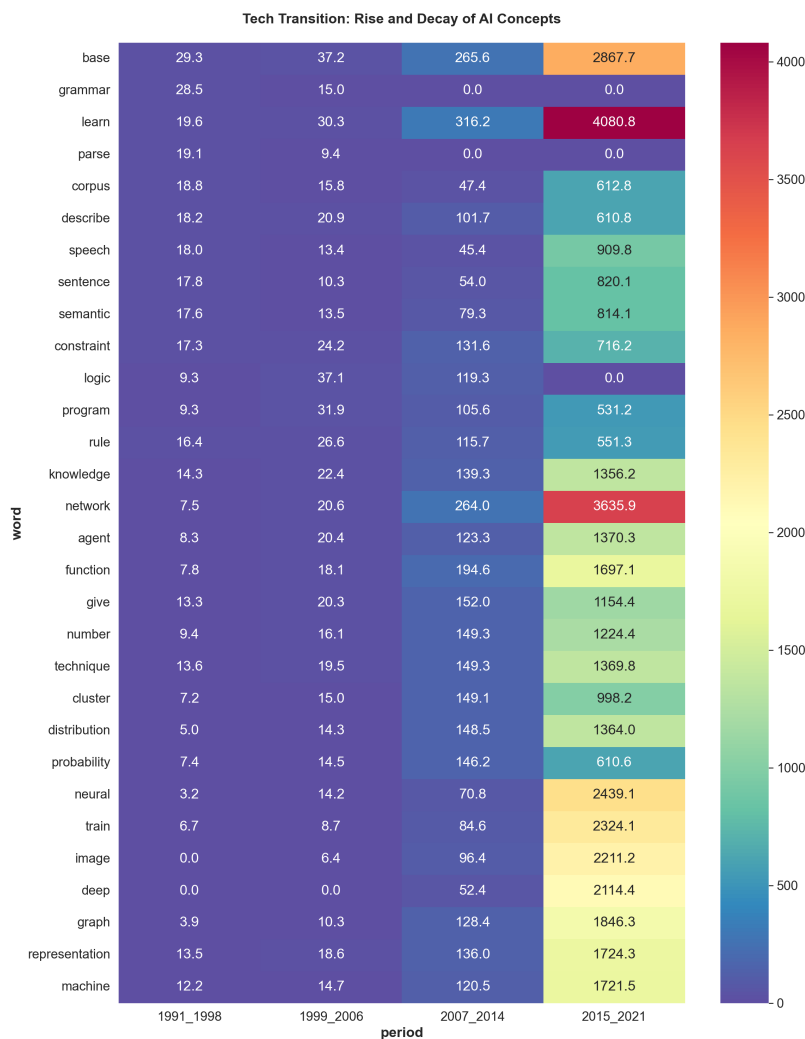


Figure 3: Rise and decay of AI concepts across periods (normalised TF-IDF). Red = high; purple = absent.

Figure 3 tracks 30 terms across all periods. Symbolic AI terms — *grammar*, *parse*, *logic* — score modestly early and disappear by Period 4. *Deep* scores 0.0 in the first two periods, rises to 52.4 in 2007–2014, and reaches 2,114.4 by 2015–2021. *Network* grows from 7.5 to 3,635.9 and *learn* from 19.6 to 4,080.8. The transition accelerates sharply after 2014, consistent with the n-gram results. Agreement between both TF-IDF methods strengthens confidence that these are genuine field-level shifts rather than noise.

3.5 Axis 2 Results: LDA Topic Analysis

We applied LDA with $k = 5$ topics to the preprocessed corpus using a bigram-aware `CountVectorizer` with a vocabulary of 1,000 features. Table 2 shows the top keywords for each discovered topic and our interpretation of each.

Table 2: LDA topic definitions (top keywords) and interpretations.

Topic	Top Keywords	Interpretation
1	learn, machine, machine learn, optimization, technique, cluster	General ML Methods
2	function, distribution, sample, loss, bound, gradient, estimate	Statistical / Learning Theory
3	agent, policy, human, environment, user, question, reinforcement	Reinforcement Learning / Agents
4	graph, label, representation, train, domain, knowledge, speech, sentence	NLP / Knowledge Representation
5	network, neural, neural network, image, deep, architecture, layer	Deep Learning / Neural Networks

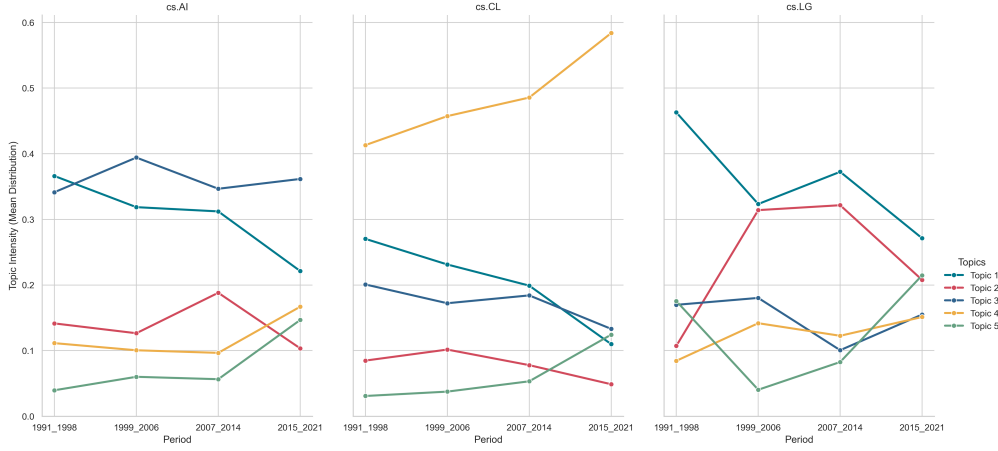


Figure 4: LDA topic intensity (mean distribution) across categories and periods. Topic 4 (orange) grows steadily in cs.CL; Topic 5 (green) rises across all subfields from 2007 onward.

Figure 4 and the underlying data reveal several clear patterns. In *cs.CL*, Topic 4 (NLP and knowledge representation) is the dominant topic throughout the entire corpus and grows consistently across all four periods, rising from an intensity of 0.41 in 1991–1998 to 0.58 by 2015–2021. This is the strongest single trend in the LDA results, reflecting the sustained and growing focus of the NLP community on representation learning, language understanding, and knowledge-grounded tasks. Topic 1 (general ML methods) correspondingly declines in *cs.CL* from 0.27 to 0.11 over the same period, as the subfield became more specialised.

In *cs.AI*, Topic 3 (reinforcement learning and agent-based systems) remains consistently prominent throughout all periods, peaking at 0.39 in 1999–2006 and staying near 0.36 by 2015–2021. This reflects the longstanding centrality of agent-based reasoning to the AI subfield, even as neural methods rose elsewhere. Topic 5 (deep learning and neural networks) grows from just 0.04 in 1991–1998 to 0.15 by 2015–2021, consistent with the broader field-wide trend.

In *cs.LG*, Topic 2 (statistical and mathematical learning theory) shows a notable spike, rising sharply from 0.11 in 1991–1998 to 0.31 by 1999–2006 and remaining elevated through 2007–2014. This aligns with the statistical learning era and the growth of kernel methods and theoretical ML research in that period. By 2015–2021, Topic 5 (neural networks) grows to 0.21, reflecting the shift of *cs.LG* toward deep learning.

These findings partially confirm our hypothesis. LDA successfully identified coherent, interpretable thematic groupings that align with known developments in each subfield. However, the dominance of the term “learn” across all topics — a consequence of it surviving the stopwords filter — reduces the discriminative sharpness of topic boundaries. This is a known limitation of LDA applied to domain-homogeneous corpora: when a field is uniformly defined by a core concept such as learning,

that concept saturates the topic-word distributions and limits the separation between topics compared to what BERTopic achieves through embedding-based clustering.

3.6 Axis 2 Results: BERTopic Evolution

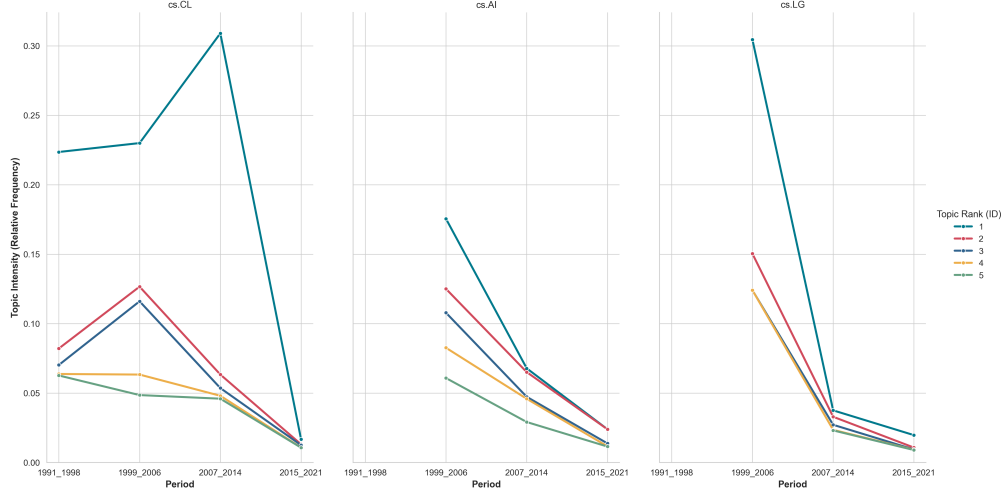


Figure 5: BERTopic topic intensity across categories and periods. Topic 1 (teal) peaks in 2007–2014 and declines sharply by 2015–2021 in all subfields.

Figure 5 shows Topic 1 dominating the early and middle periods, peaking at 0.31 in cs.CL (2007–2014) and 0.30 in cs.LG (1999–2006), before collapsing below 0.05 by 2015–2021. Rather than a failure of BERTopic, this reflects genuine field-level diversification: as publication volume expanded exponentially, research spread across computer vision, reinforcement learning, language modelling, and graph networks simultaneously. This finding complements the n-gram results and aligns with our hypothesis.

3.7 Temporal Drift: SVD and t-SNE

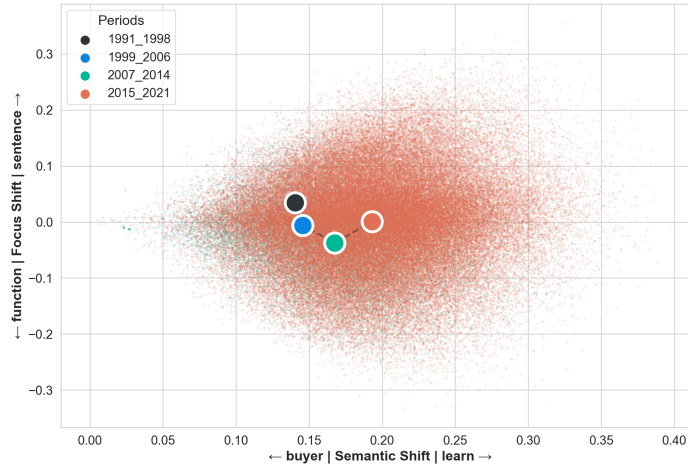


Figure 6: Truncated SVD projection. Period centroids trace a rightward trajectory along the “buyer → learn” semantic axis.

Figure 6 shows period centroids moving progressively rightward along the primary SVD axis, labelled from “buyer” toward “learn”. The trajectory is compact overall, but the largest single step occurs between Period 3 and Period 4, consistent with the accelerating shift seen in the keyword heatmap. The “learn” pole directly reflects the first principal component of 30 years of TF-IDF-weighted vocabulary variation.

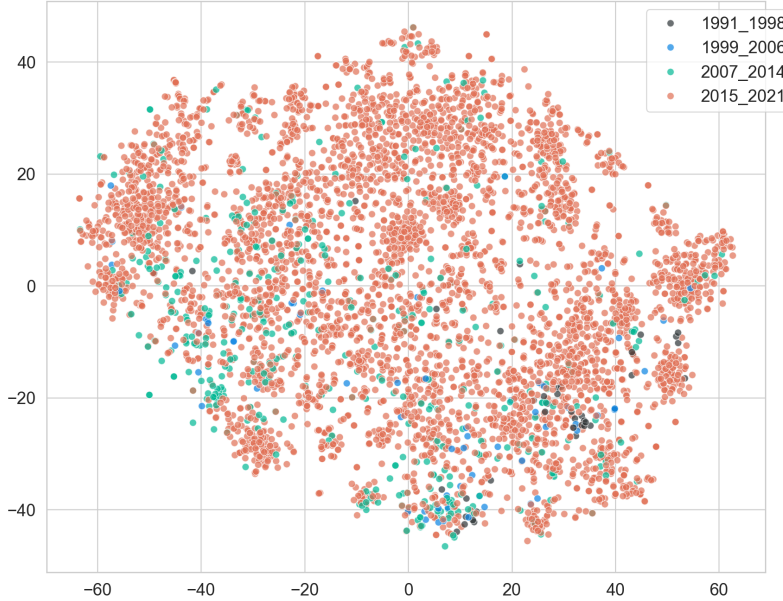


Figure 7: t-SNE of BERT embeddings (5,000 abstracts). Periods are interspersed — BERT organises by topic, not era.

Figure 7 shows BERT embeddings do not cluster by time period. A plausible explanation is that BERT encodes research content at the topic level — reinforcement learning appears in every era — rather than reflecting the vocabulary-level shifts TF-IDF captures. This is a useful negative finding: for diachronic vocabulary analysis, surface-level representations appear more sensitive to era transitions than deep contextual ones.

3.8 Error Analysis and Discussion

The custom stopwords list required manual judgement: terms like “learn” and “network” carry genuine discriminative signal but their extreme frequency in Period 4 may mask subtler thematic patterns. This directly affected LDA — the term “learn” saturates all five topic-word distributions, reducing boundary sharpness between topics. A stricter stopwords policy or larger vocabulary (> 1,000 features) would likely produce cleaner topic separation. BERTopic’s intensity collapse in 2015–2021 confirms diversification but cannot describe the content of that diversity without examining full topic label lists. The t-SNE visualisation is affected by class imbalance — Period 4 dominates the 5,000-abstract sample — meaning earlier periods are underrepresented and harder to interpret geometrically.

Despite these limitations, findings are highly consistent across methods. TF-IDF keyword and n-gram analysis produced the most interpretable and robust outputs. BERTopic added a complementary view of thematic diversification. LDA provided interpretable topics but showed less stability across hyperparameter choices. BERT embeddings were least effective for era-level temporal analysis. The main limitation of the overall approach is treating categories as static and mutually exclusive, whereas in practice many papers span subfields and period transitions are gradual.

4 Conclusions

This work applied a multi-method text analytics pipeline to 30 years of arXiv abstracts from AI, ML, and NLP subfields. Across all methods, the findings tell a consistent story: the corpus transitions from symbolic AI and formal methods in the 1990s, through probabilistic and statistical learning in the 2000s, into neural methods from 2007, and finally into a highly diverse, transformer-dominated landscape from 2015.

TF-IDF keyword and n-gram analysis were the most effective tools for detecting this transition. Multi-word phrases such as *convolutional neural*, *pre train*, and *reinforcement learn* were essential signals invisible to unigram analysis. BERTopic added a distinct insight: the modern era is characterised by the fragmentation of research into many specialised directions, reflected in sharply declining per-topic intensity after 2014. LDA surfaced interpretable thematic groupings that broadly aligned with keyword trends, though it was more sensitive to hyperparameter choices than BERTopic.

BERT embeddings did not cluster by time period, which is a useful negative result. Semantic representations encode research content at the topic level rather than the era level, making TF-IDF-based representations more appropriate for diachronic vocabulary analysis of this kind.

Open challenges include addressing class imbalance more principally, developing quantitative interpretability metrics for topic model outputs, and extending the analysis beyond abstracts to full paper texts or citation networks, which could reveal influence patterns invisible at the abstract level.

Code repository: [https://github.com/\[YOURREPOLINK\]](https://github.com/[YOURREPOLINK])

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